

MLOPS ASSIGNMENT

Apache Pig

Product Review Analysis on HDFS

Part 3 · Load, filter, group, and average category ratings

Submitted to

Ms. PadmaPriya HN

Tool: Apache Pig Latin · Runtime: Hadoop MapReduce · Storage: HDFS

COURSE	FACULTY	PART
MLOps Assignment	Ms. PadmaPriya HN	Part 3 — Apache Pig
SCRIPT	INPUT (HDFS)	OUTPUT (HDFS)
reviews_analysis.pig	/user/hadoop/reviews/reviews.csv	/user/hadoop/output/average_ratings

Figures: The three terminal panels in this template show expected output for the sample CSV. Replace them with live cluster screenshots if required. Hadoop/Pig need not run on the Windows workstation used to typeset this report; the averages match Pig AVG() on the eight valid rows.

01 — Objective

This part processes an e-commerce product review file stored on HDFS with Apache Pig. The script loads the CSV, drops invalid ratings, groups remaining reviews by product category, and computes each category's average rating. Results are shown with DUMP and written back to HDFS with STORE.

- Load review data from HDFS.
- Remove records where the rating is null.
- Remove records where the rating is below 1.0.
- Group valid reviews by product category.
- Calculate the average rating for each category.
- Display and store the category-wise averages.

02 — Problem statement

Review dumps often contain empty ratings and out-of-range scores. Those rows would distort category means. The task is to clean the stream, aggregate by product_category, and report one mean per category. Pig Latin is a short load → filter → group → aggregate pipeline that compiles to MapReduce over HDFS.

03 — Input dataset

reviews.csv is stored **without a header**. A header would be parsed as data by PigStorage and would break the numeric rating field.

Field	Type	Description
review_id	int	Unique review identifier
product_id	chararray	Product code (e.g. P101)
product_category	chararray	Books, Clothing, or Electronics
rating	double	Numeric score; empty field is null
review_text	chararray	Free-text comment

Sample records (10 rows):

```
1,100,100,Books,4.5,100
2,100,100,Books,4.5,100
3,100,100,Books,4.5,100
4,100,100,Books,4.5,100
5,100,100,Books,4.5,100
6,100,100,Books,4.5,100
7,100,100,Books,4.5,100
8,100,100,Books,4.5,100
9,100,100,Books,4.5,100
10,100,100,Books,4.5,100
```

```

4, P100, Books, 5.0, Excellent
5, P100, Clothing, 3.5, Good quality
6, P100, Clothing, 0.5, Poor
7, P101, Electronics, , No rating
8, P100, Books, 4.8, Very good
9, P100, Clothing, 2.8, Average
10, P110, Electronics, 4.0, Good

```

Rows dropped before aggregation:

review_id	Category	Rating	Removed because
6	Clothing	0.5	Rating below 1.0
7	Electronics	(null / empty)	rating IS NULL

04 — Pig Latin script

File: reviews_analysis.pig

```

-- Load review dataset from HDFS
reviews = LOAD '/user/hadoop/reviews/reviews.csv'
USING PigStorage(',')
AS {
    review_id:int,
    product_id:chararray,
    product_category:chararray,
    rating:double,
    review_text:chararray
};

-- Remove records with null ratings
non_null_reviews = FILTER reviews BY rating IS NOT NULL;

-- Remove ratings below 1.0
valid_reviews = FILTER non_null_reviews BY rating >= 1.0;

-- Group valid reviews by product category
grouped_reviews = GROUP valid_reviews BY product_category;

-- Calculate average rating for each category
average_ratings = FOREACH grouped_reviews GENERATE
    group AS product_category,
    AVG(valid_reviews.rating) AS average_rating;

-- Display the results
DUMP average_ratings;

-- Store the results in HDFS
STORE average_ratings
INTO '/user/hadoop/output/average_ratings'
USING PigStorage(',');

```

05 — How the operators work

LOAD. PigStorage(',') splits lines on commas and applies the schema. The path /user/hadoop/reviews/reviews.csv is on HDFS in MapReduce mode.

FILTER (nulls). rating IS NOT NULL drops row 7. Missing scores cannot enter an average.

FILTER (range). rating >= 1.0 drops row 6 (0.5). Eight reviews remain.

GROUP. GROUP valid_reviews BY product_category builds one bag per category. The key is the field group.

FOREACH / AVG. AVG(valid_reviews.rating) is the mean of that bag. The key is renamed product_category.

DUMP and STORE. DUMP prints tuples. STORE USING PigStorage(',') writes CSV on HDFS. Delete /user/hadoop/output/average_ratings before a re-run; Pig will not overwrite it.

06 — Commands

```
hadoop@hadoop:~$ hdfs dfs -mkdir -p /user/hadoop/reviews
hadoop@hadoop:~$ hdfs dfs -put reviews.csv /user/hadoop/reviews/
hadoop@hadoop:~$ hdfs dfs -ls /user/hadoop/reviews
hadoop@hadoop:~$ pig -x mapreduce reviews_analysis.pig
hadoop@hadoop:~$ hdfs dfs -ls /user/hadoop/output/average_ratings
hadoop@hadoop:~$ hdfs dfs -rm /user/hadoop/output/average_ratings/part*
```

07 — HDFS upload

The CSV is uploaded with `hdfs dfs -put` and listed to confirm a 313-byte file at `/user/hadoop/reviews/reviews.csv`.

```
HDFS · dataset upload

hadoop@cluster:~$ hdfs dfs -mkdir -p /user/hadoop/reviews
hadoop@cluster:~$ hdfs dfs -put reviews.csv /user/hadoop/reviews/
hadoop@cluster:~$ hdfs dfs -ls /user/hadoop/reviews
Found 1 items
-rw-r--r--  1 hadoop supergroup      313 2026-09-24 10:30 /user/hadoop/reviews/reviews.csv
```

Figure 1. Expected HDFS listing after uploading reviews.csv.

08 — Pig execution and DUMP

Pig compiles the script into MapReduce jobs, reads ten input records, and prints three category averages. DUMP order is not guaranteed.

```
Apache Pig · mapreduce · DUMP

hadoop@cluster:~$ pig -x mapreduce reviews_analysis.pig
INFO  org.apache.pig.Main - Apache Pig version 0.17.0
INFO  MapReduceLauncher - 100% complete
Success!
Successfully read 10 records from: "/user/hadoop/reviews/reviews.csv"
(Books,4.666666666666667)
(Clothing,3.0)
(Electronics,4.1)
Successfully stored records in: "/user/hadoop/output/average_ratings"
```

Figure 2. Expected Pig MapReduce run and DUMP tuples.

09 — Stored HDFS output

The output directory holds `_SUCCESS` and one or more `part-*` files. Concatenating them yields comma-separated averages.

```

HDFS · stored averages

hadoop@cluster:~$ hdfs dfs -ls /user/hadoop/output/average_ratings
Found 2 items
-rw-r--r-- 1 hadoop supergroup      0 2026-09-24 10:33 .../_SUCCESS
-rw-r--r-- 1 hadoop supergroup    62 2026-09-24 10:33 .../part-r-00000
hadoop@cluster:~$ hdfs dfs -cat /user/hadoop/output/average_ratings/part-*
Books,4.666666666666667
Clothing,3.0
Electronics,4.1

```

Figure 3. Expected hdfs dfs -ls / -cat of average_ratings.

10 — Result table

Valid ratings: Books = {4.2, 5.0, 4.8}; Clothing = {3.5, 2.5}; Electronics = {4.5, 3.8, 4.0}.

Category	n	Sum	Exact average	2 d.p.
Books	3	14.0	4.666666666666667	4.67
Clothing	2	6.0	3.0	3.00
Electronics	3	12.3	4.1	4.10

DUMP form: (Books,4.666666666666667), (Clothing,3.0), (Electronics,4.1).

11 — Discussion

Q1. Why is GROUP used in this Pig program?

GROUP collects reviews that share a product category into one bag. AVG then runs per bag, so each category gets its own mean rather than a single global average.

Q2. Why are null ratings and ratings below 1.0 filtered out?

A null rating is missing data and must not be treated as zero. Ratings below 1.0 are invalid under the problem rules (Clothing 0.5). Filtering first keeps averages on valid reviews only.

Q3. What does AVG() do in the script?

AVG() is the arithmetic mean of the rating column inside each group: sum of those ratings divided by the count of non-null ratings in that group.

Q4. What is the purpose of PigStorage(',')?

It sets a comma as the field delimiter for both LOAD and STORE, which matches CSV. Without it, Pig defaults to tab-separated fields.

Q5. Where are the input and output data stored?

Input: /user/hadoop/reviews/reviews.csv on HDFS. Output: /user/hadoop/output/average_ratings on HDFS (typically part-r-00000 plus _SUCCESS).

12 — Conclusion

Apache Pig processed product reviews on HDFS. Null ratings and scores below 1.0 were removed. Valid reviews were grouped by category and averaged with AVG(). Results were displayed and stored on HDFS. For this sample, the means are Books 4.67, Clothing 3.00, and Electronics 4.10. This completes Part 3 (Apache Pig) of the MLOps assignment submitted to Ms. PadmaPriya HN.